

## Laboratory assignment №4

**Topic:** : CRUD operations using EF Core. Data validation

**Goal:** acquire skills of implementing operations of creating, reading, updating and deleting data using ORM.

### Workflow:

### Task 1.

Using the Code First approach, create a model that, together with the existing model, will form a one-to-many relationship.

```
modelBuilder.Entity<Application>()
    .HasOne(a => a.Job)
    .WithMany()
    .HasForeignKey(a => a.JobID)
    .OnDelete(DeleteBehavior.Cascade);

modelBuilder.Entity<Application>()
    .HasOne(a => a.Candidate)
    .WithMany()
    .HasForeignKey(a => a.CandidateID)
    .OnDelete(DeleteBehavior.Cascade);
}
```

Fig 1 – Reference one-to-many with Application

### Task 2.

Implement CRUD (Create, Read, Update, Delete) operations for both models, taking into account the following requirements:

|           |      |                  |        |      |                                                  |  |  |  |                     |      |         |   |
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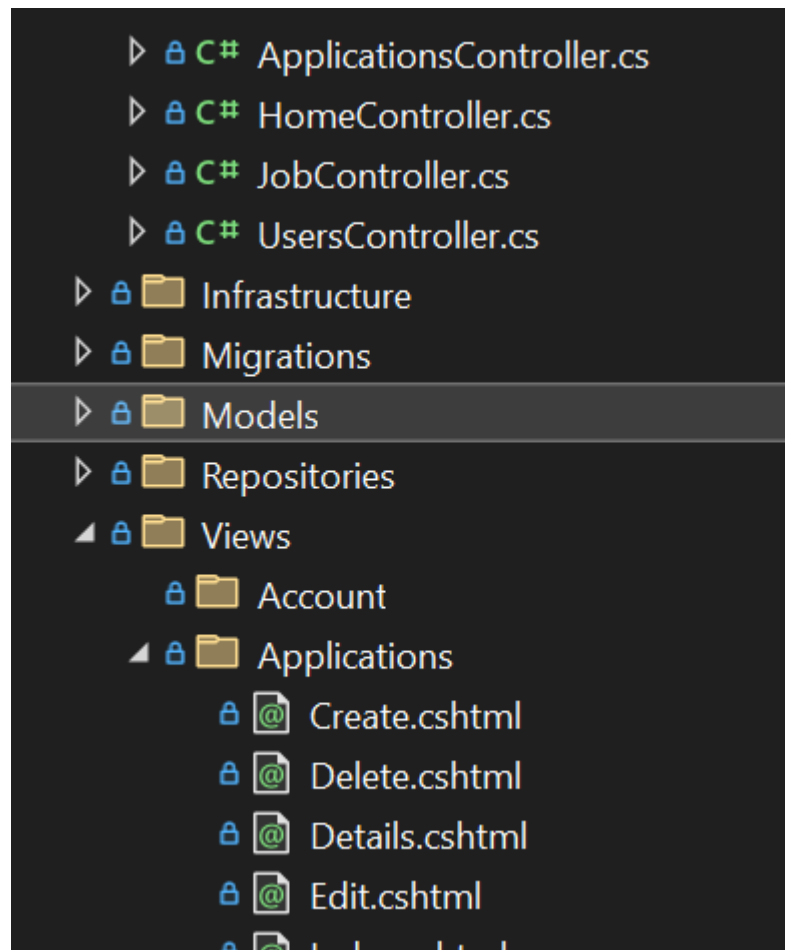


Fig 2 – Structure for CRUD-operations

- Implement a view with forms for Create and Update operations

Fig 3 – Form for Create

|      |      |                 |        |      |                                                  |      |
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Увійти

### Edit User

Name

Vadim

LastName

Lish

Login

vader

Password

1230

Fig 4 – Form for Update

- Implement a separate Details page that will display all data about the entity for Read operation

Users

Applications

## User Details

Name

Vadim

LastName

Lish

Login

vader

Phone

0981212564

City

Zhytomyr

Description

I am a student

Type

Employee

Edit

Back to List

Fig 5 – Details page

- Ask the user to confirm the deletion of the record, before performing the Delete operation

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Delete User

Are you sure you want to delete this user?

Name

Max

LastName

qwert

Login

qwert

Delete

Cancel

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Fig 6 – Delete page

## Task 3.

Add server-side validation to validate user input before performing CRUD operations. Use DataAnnotations to implement validation of input data, an example of using DataAnnotations for the Products entity is shown in the listing below:

```
[Required]
[StringLength(50)]
13 references
public string LastName { get; set; }

[Required]
[Range(1000, 9999, ErrorMessage = "Password must be a 4-digit number")]
9 references
public int Password { get; set; }

[Required]
20 references
public string Login { get; set; }

[Required]
[Phone]
12 references
public string PhoneNumber { get; set; }
```

Fig 7 – Validation with DataAnnotations (I took the Users model for display it)

**Conclusions:** During the laboratory work, I acquire skills of implementing operations of creating, reading, updating and deleting data using ORM.

|      |      |                 |        |      |                                                  |      |
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